

Problem 11

If A , B , and C are random events, simplify the expressions:

$$1^\circ (A + B)(B + C); \quad 2^\circ (A + B)(A + \bar{B}); \quad 3^\circ (A + B)(A + \bar{B})(\bar{A} + B)$$

Solution

We will use the algebra of events (Boolean algebra) to simplify these expressions.

Notation and Basic Laws

Notation:

- $A + B$ means $A \cup B$ (union: "A or B")
- AB or $A \cdot B$ means $A \cap B$ (intersection: "A and B")
- $\bar{A}$ means A^c (complement: "not A")

Basic Laws:

- Distributive: $A(B + C) = AB + AC$
- Absorption: $A + AB = A$ and $A(A + B) = A$
- Complement: $A + \bar{A} = S$ (certain event) and $A\bar{A} = \emptyset$ (impossible event)
- Idempotent: $A + A = A$ and $AA = A$
- Identity: $A + \emptyset = A$ and $AS = A$

Part 1°: Simplify $(A + B)(B + C)$

Step 1: Apply Distributive Law

$$(A + B)(B + C) = (A + B)B + (A + B)C$$

Step 2: Simplify $(A + B)B$

Using distributive law:

$$(A + B)B = AB + BB = AB + B$$

Using absorption ($B + AB = B$):

$$AB + B = B$$

Step 3: Simplify $(A + B)C$

Using distributive law:

$$(A + B)C = AC + BC$$

Step 4: Combine

$$(A + B)(B + C) = B + AC + BC$$

Using absorption ($B + BC = B$):

$$= B + AC$$

Therefore:

$$\boxed{(A + B)(B + C) = B + AC}$$

Part 2°: Simplify $(A + B)(A + \bar{B})$

Step 1: Apply Distributive Law

$$(A + B)(A + \bar{B}) = A(A + \bar{B}) + B(A + \bar{B})$$

Step 2: Simplify $A(A + \bar{B})$

Using absorption:

$$A(A + \bar{B}) = A$$

Step 3: Simplify $B(A + \bar{B})$

Using distributive law:

$$B(A + \bar{B}) = BA + B\bar{B} = AB + \emptyset = AB$$

(since $B\bar{B} = \emptyset$)

Step 4: Combine

$$(A + B)(A + \bar{B}) = A + AB$$

Using absorption ($A + AB = A$):

$$= A$$

Therefore:

$$\boxed{(A + B)(A + \bar{B}) = A}$$

Part 3°: Simplify $(A + B)(A + \bar{B})(\bar{A} + B)$

Step 1: Use Result from Part 2°

From Part 2°, we know:

$$(A + B)(A + \bar{B}) = A$$

Therefore:

$$(A + B)(A + \bar{B})(\bar{A} + B) = A(\bar{A} + B)$$

Step 2: Apply Distributive Law

$$A(\bar{A} + B) = A\bar{A} + AB$$

Step 3: Simplify $A\bar{A}$

$$A\bar{A} = \emptyset$$

Step 4: Final Result

$$A(\bar{A} + B) = \emptyset + AB = AB$$

Therefore:

$$\boxed{(A + B)(A + \bar{B})(\bar{A} + B) = AB}$$

Alternative Method for Part 3°

We can also expand all three factors:

$$(A + B)(A + \bar{B})(\bar{A} + B)$$

First expand the last two factors:

$$\begin{aligned}(A + \bar{B})(\bar{A} + B) &= A\bar{A} + AB + \bar{B}\bar{A} + \bar{B}B \\ &= \emptyset + AB + \bar{A}\bar{B} + \emptyset = AB + \bar{A}\bar{B}\end{aligned}$$

Now multiply by $(A + B)$:

$$\begin{aligned}(A + B)(AB + \bar{A}\bar{B}) &= A(AB + \bar{A}\bar{B}) + B(AB + \bar{A}\bar{B}) \\ &= A \cdot AB + A\bar{A}\bar{B} + B \cdot AB + B\bar{A}\bar{B} \\ &= AB + \emptyset + AB + \bar{A}B\bar{B} \\ &= AB + AB + \emptyset = AB\end{aligned}$$

This confirms our result.

Verification Using Venn Diagrams

For Part 1°: $(A + B)(B + C) = B + AC$

- $(A + B)$ is the region covered by A or B
- $(B + C)$ is the region covered by B or C
- Their intersection includes all of B plus the part of A that overlaps with C (but not with B)
- This is exactly $B + AC$

For Part 2°: $(A + B)(A + \bar{B}) = A$

- $(A + B)$ includes A and B
- $(A + \bar{B})$ includes A and everything outside B
- Their intersection is exactly A (since anything in both must be either in A or in both B and $\bar{B}$, which is impossible)

For Part 3°: $(A + B)(A + \bar{B})(\bar{A} + B) = AB$

- The first two factors give A
- Intersecting with $(\bar{A} + B)$ gives the part of A that is also in $(\bar{A} + B)$
- This is the part of A that is in B (since A cannot be in $\bar{A}$)
- Therefore, the result is AB

Final Answer

$$\begin{array}{ll} 1^\circ & (A + B)(B + C) = B + AC \\ 2^\circ & (A + B)(A + \bar{B}) = A \\ 3^\circ & (A + B)(A + \bar{B})(\bar{A} + B) = AB \end{array}$$

Key techniques used:

- Distributive law: $A(B + C) = AB + AC$
- Absorption law: $A + AB = A$ and $A(A + B) = A$
- Complement law: $A\bar{A} = \emptyset$ and $A + \bar{A} = S$
- Idempotent law: $AA = A$